

Size-sensitive Sequence of Tense in Hungarian

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Abstract

The English sentence *John said that Mary was pregnant* is ambiguous: Its *simultaneous* reading is *John said: ‘Mary is pregnant’*; its *backshifted* reading is *John said: ‘Mary was pregnant’*. Under Sequence of Tense Analyses of such *past-under-past* configurations (e.g. Ogihara 1995), the simultaneous reading arises if an embedded past head undergoes “deletion”-under-Agree with the matrix past head (the embedded past form is phonetically unaffected). Without deletion-under-Agree, the embedded past tense remains backshifted. The availability of deletion-under-Agree has been handled as a language-wide parameter, e.g. unlike English, Japanese lacks deletion-under-Agree (e.g. Ogihara and Sharvit 2012). In Hungarian, *speech-reporting* clauses are Japanese-like and obligatorily backshifted, whereas *non-speech-reporting* clauses are English-like, i.e. have a simultaneous reading. I propose that that speech-reporting clauses are *internally larger* and hence the impossibility of deletion-under-Agree in these clauses can be analyzed with a *Williams Cycle effect* (e.g. Williams 2003).

Keywords: Williams Cycle, size-dependent opacity, Sequence of Tense, Hungarian

1 Introduction

In English, the matrix and embedded past tenses can have a *simultaneous* reading: In (1), John can report Mary’s pregnancy at the time (i.e. *John words were: “Mary is pregnant.”*).

- (1) *John said Mary was pregnant.*

According to Sequence of Tense analyses (e.g. Ogihara 1995), the simultaneous reading arises because the *PAST* on the embedded T is “deleted” once an Agree dependency is established with the c-commanding matrix *PAST* T head as in (2). While the deleted *PAST* in (2) is phonetically unchanged, it is interpreted as a non-indexical present relative to

the matrix past tense. That is, deletion-under-Agree gets rid of the backshifting effect of *PAST*: John does not say that Mary’s pregnancy occurred *at some point prior to* “his now” but Mary’s pregnancy occurs *at* “his now”.

- (2) **Sequence of Tense rule:** $\begin{array}{c} \bullet \text{-----} \bullet \\ \vdots \quad \quad \quad \vdots \\ \dots T_{PAST} \dots [\dots T_{PAST} \dots] \end{array}$

The Sequence of Tense rule is optional, the embedded clause can remain *backshifted*; in (1), John can make a report of a pregnancy preceding “his now” (i.e. *John said: “Mary was pregnant.”*). According to Sequence of Tense approaches (for a survey see Sharvit 2020), languages like Japanese lack this Sequence of the Tense rule entirely, i.e. Japanese past-under-past structures like (3) must be backshifted (Ogihara and Sharvit 2012:p. 640).

- (3) *2005-nen ni Joseph-wa [Mary-ga (#sono-toki) zibun-o asisi-tei-ta-to]*
 2005-year at Joseph-TOP Mary-TOP that-time self-ACC love-PROG-PAST-that
sinzi-tei-ta.
 believe-PROG-PAST
 ‘Joseph to himself, in 2005: “Mary liked me”.’
 # ‘Joseph to himself, in 2005: “Mary loves me now”.’

In Section 2, I introduce the the empirical claim of the paper: Hungarian embedded clauses exhibits a *language-internal contrast* when it comes to the availability of simultaneous readings. Hungarian embedded clauses *not* encoding the content of some verbal signs (henceforth *non-speech-reporting clauses*) work like English embedded clauses like (1) in they can receive a simultaneous reading in past-under-past configurations. In contrast, Hungarian embedded clauses encoding the content of some verbal sign (henceforth *speech-reporting clauses*) work like embedded clauses in Japanese in that they do not have a simultaneous reading but must be backshifted in past-under-past configurations. That it is the *internal structure* that distinguishes these clause types is supported by the facts that there are matrix environments which can host either of these clause types and both of these clause types can serve as either arguments and adjuncts.

Following this internal-property-centered approach, Section 3 proposes that there is a *size-difference* between embedded clauses in Hungarian: While non-speech-reporting

clauses are TPs, speech-reporting clauses are SayPs, where $\text{Say} > \text{T}$ in *fseq*. Consequently, similar to other cross-clausal dependencies in Hungarian (Egressy 2023, 2025), the Agree between T heads in (2) is sensitive to syntactic size: It can be established across structurally smaller TPs but not across structurally larger SayPs. Thus, the Agree component of Sequence of Tense rule exhibits Williams Cycle effects (Williams 1974, 2003, 2013) or *selective opacity* in Agree (Keine 2016, 2019, 2020). That the Agree component of Sequence of Tense analyses fits the locality patterns of other syntactic dependencies favors these analyses for simultaneous readings over less syntactic analyses (e.g. Abusch 1988).

Section 4 derives Williams Cycle effects in Sequence of Tense by an approach relying on *size-dependent adjunction* (built on intuitions by Williams 2003; Poole 2023; Endo and Haegeman 2019; Egressy 2023; Meadows 2023): Every embedded clause adjoins to a matrix projection of its own size, e.g. TPs to TPs and SayPs to SayPs. Given that embedded *PAST* Ts can be effected as long as they are contained in a matrix TP, embedded TPs (non-speech-reporting-clauses) are adjoined to the matrix structure low enough to be affected by “deletion-under-Agree”. In contrast, SayPs adjoin too high to be affected by “deletion-under-Agree”. The paper is concluded by Section 5.

2 The two classes of Hungarian clauses

2.1 Non-speech-reporting clauses have a simultaneous reading

This subsection discusses Hungarian clauses introduced by *hogy* ‘that’ that do *not* describe the content of verbal messages. These *non-speech-reporting clauses* with the same internal structure can occur in various environments: They can be arguments of verbs of direct perception, dreaming, thought, belief, but they can also show up in non-argument positions. Independently of their matrix role (argumenthood/adjuncthood), clauses in this class have a *simultaneous* reading in past-under-past configurations. That is, non-speech-reporting clauses work like English *that*-clauses (see the Introduction and Section 3).

Hungarian embedded clauses introduced by *hogy* ‘that’ always co-occur with the

expletive *az* ‘it’, which is case-marked according to grammatical function of the embedded clause (Kenesei 1994; É. Kiss 2002). Hence, the expletive of the object clause of *lát* ‘see’ in (4) has accusative case.¹ While expletives can be *pro*-dropped if they are post-verbal, their presence is indicated by agreement and other markers on the verb (see Egressy 2023).²

The sentence in (4) exhibits *past-under-past*: The embedded past verb *volt* ‘was’ is embedded under a matrix past verb *látta* ‘saw’.³ Furthermore, (4) describes a scene of direct perception: Peti sees the event encoded in the embedded clause as it is happening at “his now”. Since this past-under-past structure can describe direct/real-time perception, we can establish that the past-under-past configuration in (4) has a *simultaneous* reading.

- (4) *Peti a saját szem-é-vel lát-t-a (az-t), [hogy Mari szomorú vol-t].*
 Peti the own eye-3SG.P-INS see-PST-3SG.DF it-ACC that Mari sad be-PST.3SG
 ‘Peti saw the following with his own eyes: “Mari is sad”.’

The behaviour of *lát* ‘see’ applies to all verbs of direct perception, hence, the complement clause of *hall* ‘hear’ also has a simultaneous reading in past-under-past configurations (for a more feasible scenario, I use a slightly different embedded sentence):⁴

- (5) *Peti a saját fül-é-vel hall-ott-a (az-t), [hogy Mari sír-t-Ø].*
 Peti the own ear-3SG.P-INS hear-PST-3SG.DF it-ACC that Mari cry-PST-3SG.INDF
 ‘Peti heard the following with his own ears: “Mari is crying”.’
 (Peti heard the noise of crying.)

Dreaming can also be handled as an example of direct perception (“seeing in a dream”). Correspondingly, the sentence in (6) with the matrix verb *álmodik* ‘dream’ has a simultaneous reading in a past-under-past configuration.

- (6) *Peti az-t álmod-t-a, [hogy Mari szomorú vol-t].*
 Peti it-ACC dream-PST-3SG.DF that Mari sad be-PST.3SG
 ‘Peti saw the following in his dream: “Mari is sad”.’

Complement clauses to verbs of thought and belief also belong to the class of non-speech-reporting clauses: It is possible to have thoughts about the events of the world without putting thoughts into a verbal message. Thus, the past-under-past sentence in (7) with matrix verbs *hisz* ‘believe’ or *gondol* ‘think’ has a simultaneous reading.

Additionally, unlike direct perception examples like (4), (5), and (6), Example (7) also has a backshifted reading: Peti can have thoughts/beliefs about Mari's sadness which occurred at some earlier point compared to "Peti's now". The backshifted reading was excluded in the direct perception examples above for pragmatic reasons: One cannot directly perceive events in the past. (Indirect perception is discussed in the next section.)

- (7) *Peti az-t {gondol-t-a/ hit-t-e}, [hogy Mari szomorú vol-t].*
 Peti it-ACC think-PST-3SG.DF believe-PST-1SG.DF that Mari sad be-PST.3SG
 'Peti {thought/believed} the following: "Mari {is/was} sad".'

Non-speech-reporting clauses can correspond to the subject of the matrix clause, too. In Example (8), since the verb *aggaszt* takes a subject clause, it is embedded with the nominative expletive *az*. However, this does not affect that, similar to the object clauses listed above, the past-under-past configuration has simultaneous reading. Furthermore, similar to (7), (8) does not involve direct perception so it has a backshifted reading, too.

- (8) *Az aggaszt-ott-a Peti-t, [hogy Mari szomorú vol-t].*
 it worry-PST-3SG.DF Peti-ACC that Mari sad be-PST.3SG
 'The following worried Peti: "Mari {is/was} sad".'

Finally, non-speech-reporting clauses do not have to be arguments. As É. Kiss (2002) notices, *hogy*-clauses having the same form as argument clauses can co-occur with intransitive matrix verbs. Consider example (9) with the intransitive predicates *irigykedik* 'be jealous' and *morog* 'growl'. These predicates can be optionally modified by a *hogy*-clause, whose expletive has the postposition 'because'. But while the embedded clause is not an argument to the matrix verbs in (9), the past-under-past configuration has a simultaneous reading, i.e. it denotes some reason occurring at the time. Since there is no direct perception, (9) has a backshifted reading, too: The reason can occur at earlier point. By the logic so far, this follows from the internal non-speech-reporting nature of the embedded clause.

- (9) *{Irigyked-t-él/ morog-t-ál} (a-miatt), hogy Mari szomorú vol-t.*
 jealous-PST-2SG.INDF growl-PST-2SG.INDF it-because that Mari sad
 'You {were jealous/ growled} because of the following: "Mari {is/was} sad".'

Finally, (10) exhibits multiple embedding: the matrix verb is the past *ámodta* ‘dreamed’, the intermediate verb is the past *látta* ‘saw’, and the deepest verb is the past *volt* ‘was’. The simultaneous reading is available between the matrix and the intermediate clause and between the intermediate and the deepest clause: Peti dreams about Mari seeing something at “Peti’s dream’s now”; Mari sees that Zsuzsi is sad at “Mari’s now”. Since there is direct perception on both levels, the backshifting is impossible on both levels.

- (10) *Peti az-t álmod-t-a, [hogy Mari a saját szem-é-vel lát-t-a*
 Peti it-ACC dream-PST-3SG.DF that Mari the own eye-3SG.P-INS see-PST-3SG.DF
(az-t), [hogy Zsuzsi szomorú volt-t]].
 it-ACC that Zsuzsi sad be-PST.3SG
 ‘Peti dreamed the following: “Mari sees the following: “Zsuzsi is sad””.’

In sum, whether the non-speech-reporting clauses of this section were object-clauses, subject-clauses, or non-argument clauses with a *because*-interpretation, they all had a simultaneous reading in past-under-past. For pragmatic reasons, direct perception examples only had a simultaneous reading, whereas other examples had a backshifted reading, too.

2.2 Speech-reporting clauses only have a backshifted reading

Embedded clause describing the content of some (verbal) sign do not have a simultaneous reading in past-under-past configurations; the embedded clause’s past must be *backshifted*. In other words, while Hungarian non-speech-reporting clauses patterned together with the embedded clauses of languages like English, Hungarian speech-reporting clauses pattern together with the embedded clauses of languages like Japanese (Ogihara and Sharvit 2012).

In (11), the matrix verbs are *mond* ‘say’, *rikolt* ‘shout’, and *morog* ‘growl’. That their embedded clauses are complement clauses is indicated by the accusative expletive *az-t* ‘it’. The embedded clauses in (11) are *speech-reporting clauses*: They describe the *content* of the verbal message conveyed by saying, shouting, or growling. The speech-reporting nature is emphasized by sentential adverbs in the embedded left-periphery, e.g. the evaluative adverb *sajnos* ‘unfortunately’ specifies Peti’s opinion of Mari’s sadness.

Example (11) exhibits past-under-past: the *hogy*-clause with the past-tense *volt* ‘was’ is embedded under the past verbs *mondta* ‘said’, *rikoltotta* ‘shouted’, and *morogta* ‘growled’. Unlike examples in the previous section, (11) has no *simultaneous* reading: Peti can only speak/shout/growl about an event that occurs before “Peti’s now”. His words must be “Mari was sad” rather than “Mari is sad”. Thus, (11) only has a *backshifted reading*.⁵

- (11) *Peti az-t {mond-t-a/ rikolt-ott-a/ morog-t-a},*
 Peti it-ACC say-PST-3SG.DF shout-PST-3SG.DF growl-PST-3SG.DF
 [*hogy (sajnos) Mari szomorú volt-t*].
 that unfortunately Mari sad be-PST.3SG
 ‘Peti {said/shouted/growled} the following: “(Unfortunately,) Mari {was/ #is} sad”.’

Notice that while *morog* ‘growl’ takes a speech-reporting argument clause in (11) expressing the *verbal message* conveyed by growling, *morog* ‘growl’ in (9) could be modified by non-speech-reporting adjunct clauses, which specified the *reason* behind the growling.

Similar to non-speech-reporting clauses, speech-reporting clauses can occur as non-argument clauses, too. For example, consider sentences (12) with the intransitive matrix predicates *telefonál* ‘call on the phone’ and *oda jön* ‘come there’. These intransitive predicates can be optionally modified by *hogy*-clauses, whose expletive is in instrumental case and which express what the matrix subject said in the phone or while approaching. In these “called/ came with the words” contexts, the embedded clauses in (12) are speech-reporting, as it is also indicated by ‘unfortunately’, which only have a backshifted reading.

- (12) {*Telefonál-t-ál/ oda jö-tt-él hoz-zám*}
 call-PST-2SG.INDF there come-PST-2SG.INDF AD-1SG.P
 (*az-zal, hogy (sajnos) Mari szomorú volt-t*).
 it-INS that unfortunately Mari sad be-PST.3SG
 ‘You {called/ approached me} (with the words: “(Unfortunately,) Mari {was/ #is} sad”.)’

Certain matrix verbs may co-occur with both non-speech-reporting and speech-reporting clauses. For example, while *hall* ‘hear’ in (5) encoded the direct perception of some event, (13) with *hall* ‘hear’ encodes that Zsuzsi heard Peti’s verbal report of Mari’s sadness, which is also facilitated by the embedded *sajnos* ‘unfortunately’. Correspondingly,

while (5) involving direct-perception had a simultaneous reading for past-under-past, (13) involving hearing someone's verbal report does not, i.e. Peti must report Mari's sadness, which occurred prior to "Peti's now". In the same way, the non-speech-reporting complement clauses of the matrix verbs *gondol* 'think' and *hisz* 'believe' had a simultaneous reading in (7). However, once the embedded clause has *sajnos* 'unfortunately' in (13), the embedded clause gets an "inner speech" interpretation and has a backshifted reading only.

- (13) *Zsuzsi {Peti-től az-t hall-ott-a/ az-t gondol-t-a/ az-t*
Zsuzsi Peti-ABL it-ACC hear-PST-3SG.DF it-ACC think-PST-3SG.DF it-ACC
hi-tt-e}, [hogy sajnos Mari szomorú vol-t].
believe-PST-3SG.DF that unfortunately Mari sad be-PST.3SG
 'Zsuzsi {heard from Peti/ thought/ believed}: 'Unfortunately, Mari {was/ #is} sad.'

Certain examples suggest that the behavior of speech-reporting clauses may be generalized to the set of clauses encoding the *content of signs*. In (14), the matrix subject does not see Mari's sadness directly but sees the signs of the sadness: the handkerchiefs left behind or the news/photos in the newspaper. The past-under-past configurations in (14) does not have a simultaneous reading: Seeing some signs, Peti thinks Mari was sad at some point prior to "his now". However, unlike the signs so far, the signs in (14) are not verbal signs. For convenience, clauses describing "the content of signs" are still called *speech-reporting*.

- (14) *Peti {az telesír-t zsebkendők-ből/ az újság-ban} lát-t-a (az-t),*
Peti the cry.in-PTCP handkerchief-PL-EL the newspaper-IN see-PST-3SG.DF it-ACC
[hogy (korá-bb-an) Mari szomorú vol-t].
that early-COMP-ADV Mari sad be-PST.3SG
 'From {the handkerchief/ the newspaper}, Peti saw: "Mari {was/ #is} sad (earlier)."

In general, perception verbs can take complements from either types. With non-speech-reporting clauses, perception verbs express *direct perception*, and only the simultaneous reading is available for pragmatic reasons: One can only directly perceive what is happening real-time. In contrast, past and future events can only be perceived via mediators such as reports, signs, photos, videos etc. In these *indirect perception* cases, perception verbs co-occur with speech-reporting clauses, which only have a backshifted reading. Matrix

predicates *not* expressing perception are only different in that with non-speech-reporting clauses, they pragmatically tolerate both the simultaneous and the backshifted reading.

In addition, (14) provides further support to the intuition that the speech-reporting nature of clauses needs to be internally defined and cannot be reduced to matrix factors like “clauses co-occurring with a matrix Agent” (cf. Pesetsky 1992): Unlike other clauses in this section, speech-reporting clauses like (14) do not co-occur with an Agent.

Speech-reporting clauses containing past-verbs in multiple embedding structures behave like sing-embeddings. In (15), the matrix clause contains the past verb *rikoltotta* ‘shouted’, the intermediate speech-reporting clause contains the past verb *morogta* ‘growled’, and the deepest speech-reporting clause contains the past verb *volt* ‘was’. There is backshifting on both levels: Peti’s shouted words must be about Mari’s growling prior to “Peti’s now”, and Mari’s growled words must be about Zsuzsi’s sadness prior to ‘Mari’s now’.

- (15) *Peti az-t rikolt-ott-a,* [*hogy Mari az-t morog-t-a,* [*hogy*
Peti it-ACC shout-PST-3SG.DF that *Mari it-ACC growl-PST-3SG.DF* that
Zsuzsi szomorú volt-t]].
Zsuzsi sad be-PST.3SG
 ‘Peti shouted saying: “Mari {growled/ #growls} saying: “Zsuzsi {was/ #is} sad”’.

In sum, speech-reporting clauses in past-under-past configurations must be backshifted. The “speech-reporting” was defined broadly: Clauses describing the content of some sign.

2.3 Multiple embedding with mixed types

In multiple embeddings, we can have different clause types: non-speech-reporting-within-speech-reporting or speech-reporting-within-non-speech-reporting. In such structures, the local behaviour of the clause types is the same as in Sections 2.1 and 2.2: Non-speech-reporting clauses have a simultaneous reading with the next lowest clause, speech-reporting clauses must be backshifted compared to the next lowest clause.

In Example (16), the matrix verb is the past-tense speech verb *rikoltotta* ‘shouted’, whose complement clause contains the past-tense *látta* ‘dreamed’. Compared to the matrix

clause, the intermediate speech-reporting clause is backshifted: Peti shouts about Mari's perception which occurred compared at some point prior to "his now". The intermediate verb *látta* '(directly) saw' takes a non-speech-reporting clause, which contains the past-tense verb *volt* 'was'. The intermediate and the deepest clause have simultaneous reading: (According to Peti), Mari directly perceived Zsuzsi's sadness while it happened. (Since the intermediate predicate is a perception verb, the backshifting is pragmatically excluded.)

- (16) *Peti az-t rikolt-ott-a, [hogy Mari a saját szem-é-vel lát-t-a*
 Peti it-ACC shout-PST-3SG.DF that Mari the own eye-3SG.P-INS see-PST-3SG.DF
(az-t), [hogy Zsuzsi szomorú volt-t]].
 it-ACC that Zsuzsi sad be-PST.3SG
 'Peti shouted saying: "Mari saw the following: "Zsuzsi is sad"'.

The opposite configuration is shown in (17): the matrix verb is the past-tense *hallotta* '(directly) heard', which takes a non-speech-reporting clause; the intermediate verb is *rikoltotta* 'shouted', which take a speech-reporting clause. In (17), the matrix subject Mari directly heard the intermediate subject Peti's shouting at "Mari's now", i.e. the matrix clause and the intermediate non-speech-reporting clause are simultaneous. (Again, perception excludes backshifting.) The intermediate clause's subject Peti shouts about some past event relative to "his now", the deepest speech-reporting clause is backshifted.

- (17) *Mari a saját fül-é-vel hall-ott-a (az-t), [hogy Peti az-t*
 Mari the own ear-3SG.P-INS hear-PST-3SG.DF it-ACC that Peti it-ACC
rikolt-t-a, [hogy Zsuzsi szomorú volt-t]].
 shout-PST-3SG.DF that Zsuzsi sad be-PST.3SG
 'Mari heard with her own ears: "Peti shouts the following: "Zsuzsi was sad"'.

Thus, similar to (16), (17) shows that only structurally adjacent clauses interact: Just because the deepest clause is contained in a non-speech-reporting clause *at some point*, the simultaneous reading is not available for the deepest past-tense predicate as its own type is speech-reporting. In short, what matters is the clause type in which past tenses are *locally* contained in. The observation that the presence of simultaneous reading solely depends on the local environment goes back to Ogiyara (1995). The English example in (18) exhibits

a past-under-*will*-under-past configuration (Ogihara 1995:p. 105). In (18), Mary's words cannot be "I am sick" even though the highest predicate *said* and the deepest predicate *was* are both past and English allows past-under-past configurations to have a simultaneous readings as in (19). Rather, English allows the simultaneous reading for past tenses iff *the next lowest tense is past*, which does not hold in (18) because the future *will* intervenes.

(18) *John said that Mary will claim that she was sick.*

(19) *Mary claimed that she was sick.*

'Mary's words were: "I am sick."' (backshifted reading set aside)

2.4 Interim summary

The availability of the simultaneous reading in Hungarian past-under-past configurations depends on the clause type. Non-speech-reporting clauses in past-under-past configurations had a simultaneous reading. Unless pragmatics excluded it in direct perception contexts, backshifting was also available for non-speech-reporting clauses. In contrast, speech-reporting clauses in past-under-past configurations did not have a simultaneous reading but were obligatorily backshifted. Thus, Hungarian non-speech-reporting clauses pattern like English embedded clauses, whereas Hungarian speech-reporting clauses pattern like Japanese embedded clauses (see the Introduction). That the two clause types are *internally* different is indicated by the facts that both types can be both argument and adjunct clauses, and there are certain matrix predicates (e.g. 'hear' and 'think') that can co-occur with either of the types. Multiple embedding examples showed that the presence of the simultaneous reading only depends on the type of the clause which locally contains the past-tense verb, i.e. simultaneous readings arise locally, as observed by Ogihara (1995).⁶

3 Size-sensitive Sequence of Tense in Hungarian

3.1 Hungarian clause-types differ in size and exhibit Williams Cycle effects

Whether an embedded clause has a simultaneous reading depends on its internal structure. In this section, I characterize this difference in internal properties as a *size-difference*:

While non-speech-reporting clauses are TPs, speech-reporting clauses are structurally larger; they have an extra SayP layer. This TP-SayP size-difference in Hungarian leads to Williams Cycle effects in cross-clausal movement, as observed by Egressy (2025).

Following Egressy (2023, 2025), I assume that Hungarian has no CP layer in its clause structure: *hogy* ‘that’ is not a C head but merely an *edge marker* (a term by Manetta 2006, 2011) adjoined to the highest projection of the embedded clause. One piece of evidence for this is shown in (20): *hogy* does not occur at the edge of an embedded clause (i.e. it is not involved in syntactic subordination), but next to an echo question. Furthermore, *hogy* is capable of occurring next to subjunctives, tensed clauses, embedded questions (Koopman and Szabolcsi 2000), or even infinitivals (Kenesei and Szeteli 2022). In short, *hogy* is not restricted to clausal embedding and does not seem to be sensitive to the category of the embedded clause in a way as English *that* can only occur next to embedded tensed declarative clauses. For a more detailed discussion, see Egressy (2023; 2025).

- (20) A: Az Ötös Lottó heti várható főnyereménye 500 millió forint.
 the five lottery weekly expected jackpot 500 million HUF
 B: *Hogy mennyi?!*
 that how.much
 A: ‘This week the estimated jackpot of Lottery 5 is 500 HUF million.’ B: ‘How much?!’

Without a CP projection headed by *hogy* ‘that’, past-tense *non-speech-reporting* clauses are bare TPs, which are headed by the $[PAST\ t_k]$ complex, whereas *hogy* is just an adjunct:

- (21) Non-speech-reporting clauses: ... V_{matrix} $[TP\ hogy\ ... [PAST\ t_k]\ ...]$

However, not all embedded clauses are TPs; I propose that speech-reporting clauses involve an extra syntactic layer on top of TP (see Kratzer 2006; Moulton 2009; Deal 2020; Krifka 2023). Syntactic layers encoding “content” or “saying” have been shown to receive an overt realization in languages like Uyghur (Major 2021) or Buryat (Bondarenko 2022). In a similar way, Deal (2020) proposes a size-difference between the complements of speech verbs and non-speech verbs to account for differences in indexical shift. While

Krifka (2023) breaks down this layer into CommitmentP, JudgementP, and ActP, other authors like Bondarenko (2022) use a single Cont(ent)P and Major (2021) uses a single SayP projection for the additional layer of speech-reporting clauses. Although I discussed in Section 2.2 that the behavior of speech-reporting may be generalized to the class of embedded clauses encoding the content of some (not necessarily verbal) signs, the ContP label would raise the question why dreams and thoughts are not encoded in ContPs.⁷ Therefore, I will be more careful and use SayP for *speech-reporting* clauses with the note that SayP can also describe the content of non-verbal signs, as depicted in (22).

(22) Speech-reporting clauses: ...V_{matrix} [_{SayP} *hogy* ...[TP ...[*PAST* *t_k*] ...]]

The presence of a SayP such that Say>T in the functional sequence can be supported by a number of Hungarian-specific arguments. Consider first (23). As discussed in Section 2, while *mond* ‘say’ takes speech-reporting complement clauses, *álmodik* ‘dream’ takes non-speech-reporting complement clauses. According to Egedi (2021) and Egressy (2023, 2025), evaluative adverbs like *sajnos* ‘unfortunately’ and *szerencsére* ‘fortunately’ occur at a fixed structural height in the Hungarian left-periphery, above foci.⁸ As indicated by (23), evaluative adverbs like *sajnos* can be hosted in the additional layer in the complement clause of *mond* ‘say’, which is a SayP (speech-reporting clause). In contrast, the TP-sized complement of *álmodik* ‘dream’ lacks the necessary structural SayP layer to host *sajnos*. Notice, furthermore, that the complement clauses of *hall* ‘hear’ and *gondol* ‘think’ are unambiguously speech-reporting in (23) because they have *sajnos*.

(23) *Pál azt {mond-t-a/ hall-ott-a/ gondol-t-a #álmod-t-a},*
Pál it.ACC say-PST-3SG.DF hear-PST-3SG.DF think-PST-3SG.DF dream-PST-3SG.DF
[_{SayP} *hogy sajnos* [TP *Ili szomorú volt*]].
that unofrutantly Ili sad be-PST.3SG.INDF
‘Pál {said/ heard someone saying/ told to himself}: “Unfortunately, Ili {is/ #was} sad.”’

The size-difference between clause types leads to differences in opacity to cross-clausal movement. In Hungarian, the TP complements of *hall* ‘hear’ and *lát* ‘see’ are transparent

to cross-clausal focusing and *wh*-movement, which both land in Spec,Foc(us)P (a.m.o. Horvath 1986; É. Kiss 2002), as illustrated in (24) and (25) (Egressy 2023, 2025). However, in (24), focusing to matrix Spec,FocP cannot proceed from the embedded clause if the embedded left-periphery has evaluative adverbs like *szerencsére* ‘fortunately’ or *sajnos* ‘unfortunately’ (Egressy 2023, 2025), i.e. under the current analysis, if the clause is not a TP but a SayP. Similarly, if the non-speech-reporting (TP) complement of *lát* ‘see’ is replaced with the speech-reporting (SayP) complement of *morog* ‘growl’ in (25), the embedded clause is opaque to *wh*-movement to the matrix Spec,FocP (Egressy 2023:Footnote 9).⁹

- (24) { *Szerencsére/ sajnos* } [(CSAK) A VÁROS-ON]₁ *hall-ott-am*,
 fortunately unfortunately only the city-SUP hear-PST-1SG.DF
 [*hogy (??szerencsére/ ??sajnos) keresztül fut-ott-ak* ____₁].
 that fortunately unfortunately across run-PST-3PL.INDF
 ‘(Un)fortunately, I heard that (??(un)fortunately) they ran across ONLY THE CITY.’
- (25) [*Melyik város-on*]₁ { *lát-t-ad/ ??morog-t-ad* }, [*hogy keresztül fut-ott-ak* ____₁]?
 which city-SUP see-PST-2SG.DF growl-PST-2SG.DF that across run-PST-3PL.INDF
 ‘Which city did you {see/ ??growl} that they ran across?’

The opacity-differences between Hungarian TPs and SayPs fit the cross-linguistic pattern of Williams Cycle (Williams 1974, 2003, 2013) defined in (26), which is handled here as a descriptive generalization. The most common illustration of the Williams Cycle is the ban on *hyperraising* in English: While the TP in (27) is transparent to raising to Spec,TP, the CP in (28) is not (Chomsky 1973). This fits the Williams Cycle in (26) given that C>T in the *f*(unctional) *seq*(uence). Williams Cycle effects have been identified in a number of other languages, e.g. German (Müller 2014a,b), Swahili (Meadows 2023), Georgian (Bondarenko 2024), Mandarin Chinese (Meadows and Yan 2024).

- (26) **Williams Cycle:** Movement to Spec, β P cannot proceed from Spec, α P or across α P, if $\alpha > \beta$ in some functional sequence. (based on Williams 2003; Poole 2023)

- (27) [TP Ed₁ seems [TP ____₁ to be sad]] (28) *[TP Ed₁ seems [CP ____₁ is sad]]

As in the matrix clause of (24), evaluative adverbs precede foci and *wh*-phrases in the Hungarian left-periphery (Egedi 2021). Therefore, Egressy (2023, 2025) assumes that the projection hosting evaluative adverbs, which is called here SayP, dominates the FocP projection hosting foci and *wh*-words. Both SayP and FocP dominate TP, where tense is encoded. Thus, under the functional sequence Say>Foc>T, the ban on focusing and *wh*-movement to Spec,FocP from SayP falls under (26). That Hungarian cross-clausal movement also obeys the Williams Cycle in general is discussed by Egressy (2023).

3.2 The Sequence of Tense analyses of simultaneous readings

The English past-under-past configuration in (29) is ambiguous: Under the *backshifted* reading, John’s words were “Mary was pregnant”, under the *simultaneous* reading, John’s words were “Mary is pregnant” (e.g. Abusch 1988, 1997; Ogihara 1995). Sequence of Tense analyses (among others, Ogihara 1995) provide one possible derivation for simultaneous readings: After Agree holds between matrix and the embedded Ts, the embedded *PAST* is not interpreted at LF. The main theoretical claim of this paper is that Sequence of Tense analyses get extra support from Hungarian: Once we extend Williams Cycle effects to Agree, the SOT-Agree between T heads patterns like other syntactic operations in Hungarian in that it obeys the Williams Cycle. While this subsection introduces Sequence of Tense analyses, the next one defines Williams Cycle in Agree, and shows how Sequence of Tense analyses and not other analyses are favored by Williams Cycle effects in Hungarian.

(29) *John said Mary was pregnant.*

Sequence of Tense analyses (e.g. Ogihara 1995) take the backshifted reading of past-under-past configurations like (29) to be the default. The backshifted LF of (29) is provided in (30). Following the assumptions by Sharvit (2020), $[PAST\ t_k]$ complexes are handled as quantifiers with the meaning “some time prior to t_k ”, which need to be raised to the edge of their respective clauses. Quantifier raising leaves behind t_j variables, which are of the type i , a type for times. Hence, given that the matrix evaluation time is t_0 , which is “our

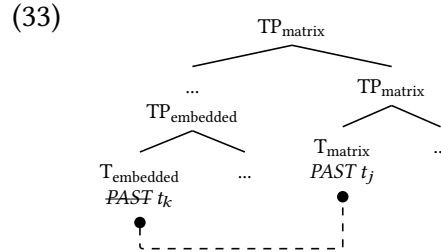
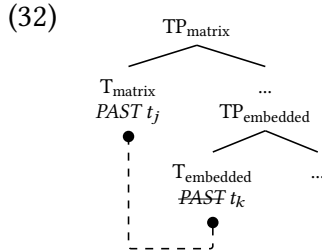
now", the quantifier $[PAST\ t_0]$ means "some time preceding our now" in (30). $[PAST\ t_0]$ at the edge of the matrix clause binds John's speaking time t_1 in $[say\ t_1]$. Furthermore, "John's now" is t_2 , which is the embedded evaluation time. Thus, $[PAST\ t_2]$ is "some time time prior to John's now", which binds Mary's pregnancy time t_3 . In short, John's speaking time is some past time relative to "our now", and the pregnancy happens at some past time *backshifted* relative to "John's now". By and large, the pregnancy is past-in-the-past.

(30) $[PAST\ t_0]\ \lambda_1\ [John\ [say\ t_1]\ \lambda_2\ [[PAST\ t_2]\ \lambda_3\ [Mary\ [be\ t_3]\ pregnant]]]$

Sequence of Tense accounts (e.g. Ogihara 1995) derive the simultaneous reading by saying: "[A] tense that is c-commanded by an agreeing tense (past-under-past, present-under-present) may optionally be converted into a zero-tense (i.e., equivalent to a relative present tense)" (Ogihara and Sharvit 2012:p.646). The formalization used here is (31).

(31) **The Sequence of Tense (SOT) Rule:**

At any point in the syntactic derivation, if a projection of $T_{[PAST\ t_j]}$ contains another $T_{[PAST\ t_k]}$, these Ts or TPs may Agree, after which *PAST* in the contained $T_{[PAST\ t_k]}$ is not interpreted at LF (marked as $T_{[PAST\ t_k]}$), i.e. it becomes a non-indexical present.



For the purposes of the paper, the rule in (31) must exhibit three crucial properties; beyond these, there is some room for variation in theoretical assumptions. The first fixed point is that given a T has a TP projection that *contains* the another T, the contained T is affected. That is, in the spirit of Cyclic Agree (Rezac 2003; Béjar and Rezac 2009; Clem 2022), the operation is not limited to configurations with T heads c-commanding other T heads, as in (32), but (31) can apply as long as the matrix T projects, i.e. even in configurations like (33). While Ogihara (1995) used a similar intuition to allow the SOT rule to affect embedded T heads occurring within the matrix Spec,TP (i.e. in sentential

subjects), the containment-based approach will allow the SOT rule to affect T heads in clauses adjoined to the matrix TP in Section 4. (For CED-compatibility, see Footnote 11.)

The second fixed point is that the rule involves a *syntactic* dependency between two constituents of the category T. Once this dependency is established, the *interpretation* of the “contained T” is special in that it does not have the backshifting effect of genuine *PAST* Ts, even though its *phonetic realization* is still the same as “genuine/backshifting” *PAST* Ts. The *PAST*-notation in (31) follows the tradition that describes the LF-affect as the “deletion-under-Agree” of the embedded *PAST*, after which the embedded T is “present-in-past-disguise” (Ogihara and Sharvit 2012; Sharvit 2020). That is, T starts out as *PAST* but loses the backshifting semantics. One alternative to the “deletion-under-Agree” could come from the “feature-transmission-under-Agree” analysis of fake indexicals (Kratzer 2009; Wurmbrand 2017), which could be implemented as follows: If the embedded T head is a feature-less variable t_k (non-indexical present), it can Agree with the matrix T in a way that it acquires the phonetic shape of past morphemes but it remains t_k at LF. A third alternative can be to handle SOT in the fashion of dependent case (Marantz 1991): If two *PAST*s occur in a TP, the one that does not project the TP is not interpreted at LF. As long as the non-backshifting interpretation of a phonetically past morpheme is achieved via a syntactic dependency between Ts and TPs, either of these three options could work for the purposes of this paper. For convenience, I will keep using the deletion-under-Agree-notation.

The final fixed point is that (31) is an optional rule which can only apply at *Narrow Syntax*. After the containment holds and Agree is established, the contained *PAST* is not handed to LF. This contrasts with the deletion-at-LF by Ogihara (1995), but it ensures that *PAST* t_k is not QRd vacuously and, more importantly, the SOT rule is constrained by *syntactic locality*. While the feature-transmission-under-Agree captures the phonetic *PAST*-realization of the embedded T, I follow the traditional “SOT-deletion” notation and say that (31) makes *PAST*s uninterpreted at LF without changing the phonetic form.

In the simultaneous LF of (29) in (34), the $[PAST\ t_2]$ is not a quantifier, it is a non-indexical present t_2 in the embedded T. This gets bound by the embedded evaluation time “John’s now”, i.e. John reports that Mary’s pregnancy occurs at “his now”.

- (34) $[PAST\ t_0]\ \lambda_1\ [John\ [say\ t_1]\ \lambda_2\ [Mary\ [be\ [PAST\ t_2]]\ pregnant]]$ becomes
 $[PAST\ t_0]\ \lambda_1\ [John\ [say\ t_1]\ \lambda_2\ [Mary\ [be\ [t_2]]\ pregnant]]$

3.3 Sequence of Tense exhibits Williams Cycle in Agree in Hungarian

In this section I propose that the Hungarian-internal differences in the availability of simultaneous readings arise because the syntactic dependency required by the SOT rule is sensitive to syntactic size. More specifically, SOT obeys the Williams Cycle (Williams 1974, 2003, 2013) just like other Hungarian syntactic dependencies did in Section 3.1.

Selective opacity or Williams Cycle effects are not limited to movement, but may occur in other syntactic dependencies like Agree (Keine 2016, 2019, 2020) and Case (Poole 2023). For example, Keine (2020) claims that Long Distance Agreement in Hindi is sensitive to size: The ϕ -probe, which is located on matrix T, can find its DP goal if it is contained in a νP clause, but not if it is contained in a CP clause (for examples and details, see Keine 2020). This can be analyzed as Williams Cycle in Agree in (35): Under the *fseq* $C > T > \nu$, the ϕ -probe on T is blocked by CPs because $C > T$ in *fseq* but not by νP s because $T > \nu$ in *fseq*.

- (35) **Williams Cycle in Agree:** No Agree from β or βP can search in αP , if $\alpha > \beta$ in *fseq*.

The SOT rule in (31) requires a syntactic Agree dependency between the embedded T head and a matrix T head or TP projection. In (36), this Agree dependency can cross TP boundaries, so the *PAST* operators on the T heads of TPs can be affected/“deleted” by the SOT rule. This is why Hungarian non-speech-reporting clauses, which were analyzed as TPs in Section 3.1, may have a simultaneous reading. (The SOT-deletion is optional, so non-speech-reporting clauses may be still backshifted.) In contrast, the Agree dependency cannot cross SayP boundaries, as depicted in (37). Hence, *PAST* in speech-reporting

clauses, which are SayPs, cannot be affected/"deleted" by the SOT rule. This is why non-speech-reporting clauses do not have a simultaneous reading but must be backshifted.

$$(36) \begin{array}{c} \bullet \text{-----} \bullet \\ \text{[TP T}_{[PAST\ t_j]} \dots [\text{TP T}_{[PAST\ t_k]} \dots]] \end{array} \quad (37) \begin{array}{c} \bullet \text{-----} \times \\ \text{[TP T}_{[PAST\ t_j]} \dots [\text{SayP} \dots [\text{TP T}_{[PAST\ t_k]} \dots]] \end{array}$$

The contrast between (36) and (37) fits the Williams Cycle pattern in (35): Since Say>T in *fseq*, the dependency between Ts and TPs cannot cross SayPs but can cross TPs. That is, the size-sensitivity of the SOT rule fits the size-sensitivity of other syntactic operations in Hungarian, e.g. in (24) and (25), while TPs are transparent to movement to Spec,FocP, SayPs are opaque because Say>Foc>T in *fseq* (for further examples, see Egressy 2023).

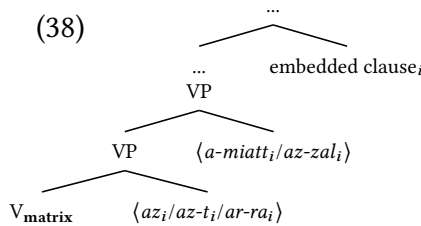
Williams Cycle-style sensitivity to syntactic size favors the SOT analyses over other analyzes for simultaneous readings . The most dominant alternative to SOT analyses are the *de re* approaches (e.g. Abusch 1988; Heim 1994). Under certain *de re* approaches (Heim 1994; Deal 2018), simultaneous readings may be derived by the “*res*-movement” of *PAST* from embedded clauses to an A-position internal to a matrix VP. One could argue that such *res*-movement to the matrix VP could be constrained in terms of syntactic locality: It can proceed from TPs (non-speech-reporting clauses), but not from SayPs (speech-reporting clauses). While this pattern is derivable by less restrictive theories of selective opacity (e.g. Keine 2016, 2019, 2020), it violates the Williams Cycle in (26) given that *fseq* is Say>T>V. This is critical because considering Hungarian in general, syntactic operations exhibit a more restrictive, Williams Cycle-style pattern of locality (Egressy 2023), that is, *res*-movement would be exceptionally free. In contrast, the SOT-rule-based approach involves a dependency between Ts or TPs and obeys the Williams Cycle.¹⁰

4 A size-dependent adjunction account for Hungarian and beyond

Williams Cycle effects in Sequence of Tense can be accounted for by various theories capable of deriving selective opacity in Agree. While one of the most well known accounts of selective opacity is Keine’s *horizon-based account* (Keine 2016, 2019, 2020), this account

is capable of deriving patterns which are excluded by the Williams Cycle. Since Sections 3.1 and 3.3 have shown that movement and Agree in Hungarian fit the Williams Cycle, i.e. a more restrictive pattern than the horizons account would predict. Hence, *size-dependent accessibility*, specifically developed for Williams Cycle effects, offer a more restrictive and therefore better alternative. In accounts relying on *delayed* or *size-dependent accessibility* (Williams 2003; Poole 2023; Meadows 2023; Egressy 2023; Bondarenko 2024), the internal structure of embedded clauses is not accessible from the matrix clause before the matrix clause is built up to their size. That is, the internal structure of larger clauses is accessible later. Building on the properties of Hungarian and the cross-linguistic variation, the specific implementation used here involves the *size-dependent high adjunction* of embedded clauses. The end of the section addresses how cross-linguistic SOT-variation may be derived.

Following É. Kiss (2023), I assume high-adjunction for all *hogy*-clauses: While the expletive *az* ‘it’ is in a VP-internal argument or adjunct position (suffixed with the right case or P), the embedded clause associated with it is high-adjoined, as in (38). In the mode of combination of *hogy*-clauses, the high-adjunction account does not have an argument vs. adjunct distinction; all *hogy*-clauses are high-adjoined. The argument vs. adjunct distinction only exist for expletives. This is a welcome result because the availability of the simultaneous reading did not depend on the argument/adjunct status of the clause in Section 2. The structure in (38) is supported by (39): Hungarian *hogy*-clauses cannot occur VP-internally (e.g. É. Kiss 2002) but must occur in the right (or left) periphery, to right of adjuncts, where they are linearly separated from their expletives and matrix verbs.

- (38) 
- (39) *Mari hivatkoz-ott* {ar-ra_i/ Ø-ra_i} < ma>,
 Mari refer-PST.3SG.INDF it-SUB pro-SUB today
 [*hogy Peti tegnap sír-t-Ø*]_i < *ma>.
 that Peti yesterday cry-PST-3SG.INDF today
 ‘Mari referred this today: “Peti cried yesterday.”’

To ensure that we get the right semantic interpretation, I assume (39). For example, complement clauses are substituted into the VP-internal position of the accusative *az-t*.

(40) Every *hogy*-clause is substituted into the position of its expletive at LF.

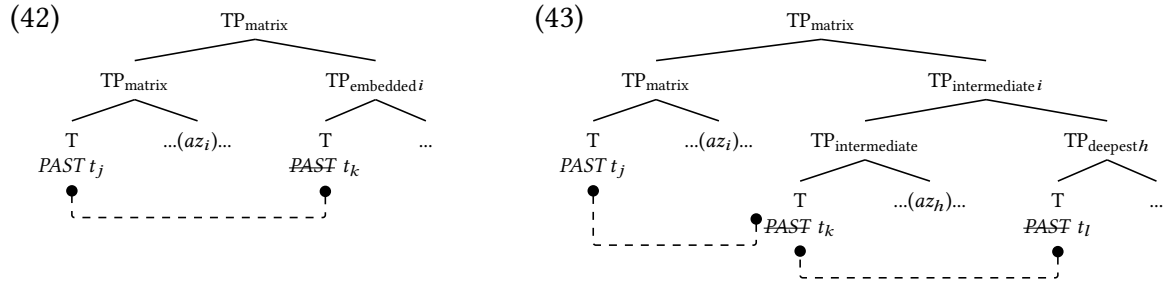
According to Williams' Level Embedding Conjecture (Williams 2003; Poole 2023), an αP can only be embedded in a matrix clause that is also built up to αP ; e.g. embedded CPs occur in the matrix VP when the matrix C merges in. This delayed accessibility intuition is implemented by Meadows (2023) as a theory on *adjunction timing*: Larger argument (and adjunct) clauses adjoin to the matrix clause later in the derivation, but the adjunction site of any clauses can be still low, it does not have to extend the matrix root node (i.e it can be countercyclic). For example, a CP adjoins within the matrix VP when the matrix C merges in. Building on the obligatory VP-external and peripheral position of Hungarian clauses in structures like (39), the constraint in (41) implements Meadows' (2023) adjunction-timing idea in *cyclic* way: Every clause must adjoin to the matrix projection that aligns with its size (as in Endo and Haegeman 2019). Unlike in other delayed-accessibility accounts (Williams 2003; Poole 2023; Meadows 2023), the embedding of clauses is cyclic under (41) because the bottom-up derivation is not violated by going back the lower layers (e.g. VP) of the matrix structure: The adjunction of the embedded clauses happens at the root node.

(41) **Size-dependent adjunction:** αP must adjoin to a structure that is built up to αP .

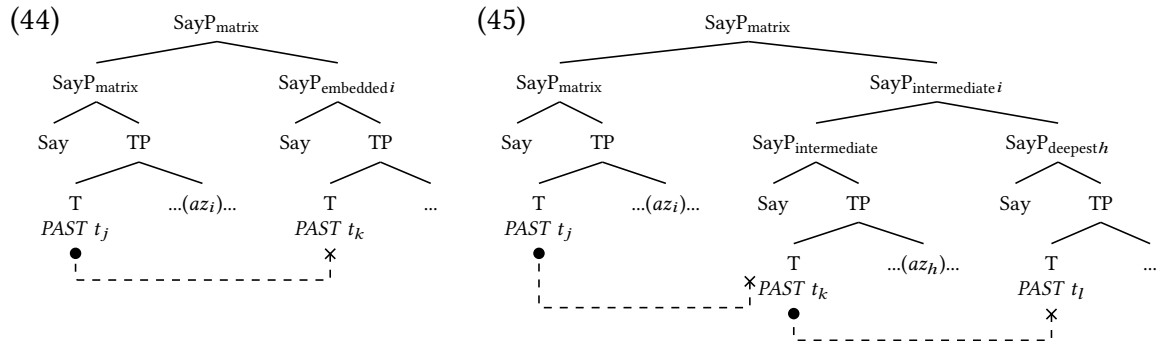
A cyclic alternative to (41) is the size-sensitive extraposition account by Egressy (2023). However, space does not permit to discuss the assumptions needed for this account.

Under (41), TPs (non-speech-reporting clauses) are adjoined to the matrix TP, as in (42). Recall that the SOT-definition in (31) followed the spirit Cyclic Agree (Rezac 2003; Béjar and Rezac 2009; Clem 2022), and allowed Agree in *Narrow Syntax* between T heads as long as a matrix TP projection contained the embedded T. This is the case in (42): The highest matrix TP projection contains the embedded T with *PAST*, so deletion-under-Agree is possible. Once the embedded clause does not have *PAST* at LF and embedded clauses

are substituted in the VP-internal position of the expletive by (40), a *simultaneous* reading arises for non-speech-reporting clauses (TPs).¹¹ In the multi-layered embedding of non-speech-reporting clauses like (10), the pattern is the same: Every embedded TP adjoins to the next lowest clause's TP, as in (43). Since the deepest *PAST* is contained in the highest TP projection of the intermediate clause, and this TP is a projection of the intermediate *PAST*, the deepest *PAST* is deleted. After this, the intermediate *PAST* also gets deleted, once the intermediate TP adjoins to the matrix TP because the highest matrix TP is a projection of the matrix *PAST*. In short, the simultaneous reading is derived in both cases.¹²



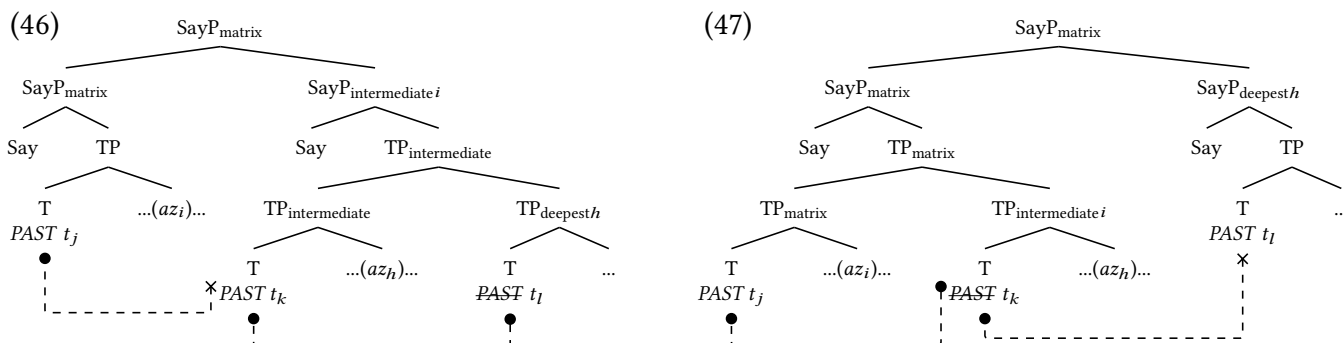
In contrast, embedded SayPs (speech-reporting clauses) adjoin to the matrix SayP, which is too high to be contained by a matrix TP projection, as in (44). Consequently, the embedded *PAST* cannot be deleted before LF in (44). Since the SOT rule is restricted to Narrow Syntax, the embedded *PAST* cannot be deleted after SayP gets substituted in the position of its expletive at LF, the embedded SayP has a *backshifted* reading.



The multi-layered embedding of speech-reporting clauses in (15) works in the same way. In (45), since the deepest *PAST* is not contained in either the intermediate or the matrix TP,

and the intermediate *PAST* is not contained in the matrix TP, the SOT rule cannot apply. So the deepest *PAST* backshifts the deepest clause relative to the intermediate clause and the intermediate *PAST* backshifts the intermediate clause relative to the matrix clause.

In Example (16) (Section 2.3), the deepest clause is a non-speech-reporting clause (TP) and the intermediate clause is a speech-reporting clause (SayP). Such configurations are depicted in (46): While the deepest TP adjoins to the intermediate clause's TP projection, the intermediate SayP adjoins to the matrix clause's SayP projection. Since the deepest *PAST* is contained in an intermediate TP, the SOT-deletion of the deepest *PAST* is possible. Therefore, the deepest clause can be simultaneous with the intermediate clause, as expected. The intermediate *PAST* is not contained in the matrix TP because the intermediate SayP attaches too high, so the intermediate clause is backshifted relative to the matrix clause.



The opposite configuration is Example (17) (Section 2.3), which is depicted by (47): Here the intermediate clause is a non-speech-reporting TP, whereas the deepest clause is a speech-reporting SayP. Under (41), the intermediate TP adjoins to the matrix TP. Since the highest matrix TP contains the intermediate *PAST*, the SOT can apply, so the intermediate clause can be simultaneous with the matrix clause. Since the intermediate clause does not have a SayP projection to which the deepest SayP could adjoin, the deepest SayP adjoins to the matrix SayP. Since the deepest *PAST* is contained in neither the matrix nor the intermediate TP, the deepest *PAST* cannot undergo SOT-deletion. At LF, the embedded clauses are substituted in the position of their expletives; this is how the deepest clause is

backshifted relative to the intermediate clause. This derives Ogihara’s generalization discussed in Section 2.3: If the SOT-deletion of the deepest *PAST* is impossible from the intermediate clause, it will be impossible from the matrix clause, too.

The cross-linguistic variation mentioned in the Introduction requires further investigation. One option is to derive the difference between the size-sensitive SOT in languages like Hungarian and the size-insensitive SOT in languages like English from difference in attachment-heights.¹³ Under my account, Hungarian cross-clausal syntactic operations, including the SOT’s deletion-under-Agree, are size-sensitive as Hungarian obeys (41) and has size-sensitive adjunction for *hogy*-clauses. In contrast, in English, the Agree dependency between Ts or TPs violates the Williams Cycle in Agree in (35) assuming English past-tense clauses are CP-sized and $C > T$ in *fseq*. This size-insensitivity in English SOT can be derived if the size-dependent adjunction in (41) does not hold in English. If English CPs occur in a position contained in TP (e.g. in a V-sister position), the Agree between Ts or TPs can hold irrespectively of their size, i.e. the SOT-rule can delete *PAST* in CPs.

This leaves us with two main open questions. Firstly, the ban on hyperraising from English CPs to Spec,TP discussed in Section 3.1 needs to be derived independently (e.g. from binding, as in Grishin 2025). Secondly, it is not clear whether SOT-variation can be *completely* derived from variation in adjunction-height, e.g. whether Japanese can be argued to lack simultaneous readings because its tensed, CP-sized complements are high-adjoined, similar to its CP-sized adjuncts (as argued by Endo and Haegeman 2019).

5 Conclusion

In this paper, I demonstrated that in Hungarian, non-speech-reporting embedded clauses (TPs) can have a *simultaneous* reading in past-under-past configurations, whereas speech-reporting embedded clauses (SayPs) only have a *backshifted* reading. Given that the SOT rule requires Agree between Ts or TPs, the (im)possibility of *PAST*-deletion was described

as a Williams Cycle effect (e.g. Williams 2003) in Agree (e.g. Keine 2016). In my account, I assumed that all embedded clauses adjoin high, when the matrix clause reaches their size: TPs to TPs and SayPs to SayPs. I assumed furthermore that the SOT rule can apply as long as a projection of the matrix *PAST* contains the embedded *PAST*. This rule could apply to TP-TP adjunction structures but not to SayP-SayP adjunction structures, thus the embedded *PAST* could only be deleted in the former case. Ogiwara's generalization was also derived from the idea that SayP-sized embedded clauses can never adjoin low enough to be affected by the SOT rule. This strongly syntactic analysis is capable of handling the intra-linguistic variation of Hungarian past-under-past configurations, whereas previous accounts handled the availability of simultaneous reading as a language-wide parameter.

In general, that the Agree of the SOT rule obeys constraints of syntactic locality, here the Williams Cycle, provides a special support for accounts making use of the SOT rule.

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Notes

¹ The examples follow Leipzig Glossing rules. The abbreviations: 1 = first person, 2 = second person, 3 = third person, ABL = ablative, ACC = accusative, ADV = adverb, COMP = comparative, DF = definite, EL = elative, IN = inessive, INDF = indefinite, INF = infinitive, INS = instrumental, PL = plural, P = possessive, PRS = present, PST = past, PTCP = participial, SG = singular, SUB = sublative, SUBJ = subjunctive. The uncited Hungarian data comes from my own data collection. Although I am a native speaker of Hungarian, the grammaticality of critical sentences were double-checked with native speaker linguists and in a survey 15 native speakers (conducted Summer, 2022). The data represents the standard dialect spoken in Budapest.

² If the expletive is not marked by nominative/accusative (i.e. a structural/non-P-like case), but the clause is still an argument, the expletive cannot be fully *pro*-dropped because the case suffix (or P morpheme) needs to be present, as discussed by Marácz (1989); Dékány and Hegedűs (2021). For example, the remnant of the sublative case suffix needs to be present in the argument-embedding example in (39) (Egressy 2023). In contrast, adjunct clauses like (9) and (12) are never embedded with nominative/accusative expletives, and their expletives can be fully dropped. É. Kiss (2002) refers to adjunct *hogy*-clauses as *loose adjunction*.

³ Following Stowell (2008), I use stative embedded predicates whenever possible.

⁴ Both clause types can stand with factive or non-factive matrix predicates. While the embedded clauses in (4) and (5) were factive, this does not have to be the case for verbs of direct perception (Kiefer 1978). The pronominal *úgy* introduces the possibility of misperception in (48), while (48) has a simultaneous reading just like (4) does. While the main text in Section 2.2 has only non-factive predicates with speech-reporting clauses, e.g. we can add “But she was not sad” to (11), by the removal of the expletive *azt* ‘it’, the matrix predicate *mondta* ‘say’ becomes factive (Vendler 1980; Kiefer 1986; Tóth 1999; Abrusán 2025). However, the factivity-switch does not change that speech-reporting clauses have only a backshifted reading. Therefore, I follow Abrusán (2025) by stating that factivity (and its sub-components) depend on the matrix position of the embedded clause, and not on its internal structure De Cuba (2007).

- (48) *Peti a saját szem-é-vel úgy lát-t-a (az-t), [hogy Mari szomorú vol-t].*
 Peti the own eye-3SG.P-INS so see-PST-3SG.DF it-ACC that Mari sad be-PST.3SG
(Pedig vidám vol-t)
 but happy be-PST.3SG
 ‘Peti (potentially incorrectly) saw with his own eyes: “Mari is sad.” (But she was happy.)’

⁵ To test further the availability of the simultaneous reading, one can add ‘at 9:55 am’ in the matrix clause and ‘then/in that moment’ to the embedded clause: If the latter time cannot refer to the former time, the simultaneous reading is not available. The two Hungarian speakers I asked confirmed that this test shows that only non-speech-reporting clauses have simultaneous readings.

⁶The paper does not discuss present-under-past configurations. For most cases, the embedded present in Hungarian is be non-indexical, i.e. it does not access the “our now”. Therefore, (49) is felicitous with *two years ago* (i.e. Mary’s pregnancy does not need to last for two years). For the non-speech-reporting clauses above, present-under-past is the only way encode the simultaneous reading; for non-speech-reporting clauses, the present-under-past and the past-under-past configurations both encode simultaneous readings. Probably, this is why many Hungarian speakers prefer to use the unambiguous present-under-past configurations to express the simultaneous reading of non-speech-reporting clauses; (Ogihara and Sharvit 2012) notes that Hebrew has similar patterns. To my knowledge, the only instances of indexical present occur in the complement clauses of verbs like *elér* ‘achieve’ or *elintéz* ‘arrange’. In (50), *two years ago* is infelicitous for anyone who knows football: It is never the case that single goal actively keeps the team on the top of the billboard for two seasons. *Elér* and *elintéz* are special as they can take both subjunctive or indicative clauses.

- (49) *(Két évvel ezelőtt) Peti {lát-t-a/ az-t rikolt-ott-a}, [hogy Mari terhes Ø].*
 two years ago Peti see-PST-3SG.DF it-ACC shout-PST-3SG.DF that Mari pregnant be.PRS.3SG
 ‘Two years ago, Peti {saw with his own eyes/ shouted}: “Mari is {sad/pregnant}”’.
- (50) *(#Két évvel ezelőtt) Szoboszlai elér-t-e, [hogy a Liverpool az első Ø].*
 two years ago Szoboszlai achieve-PST-3SG.DF that the Liverpool the first be.PRS.3SG.INDF
 ‘(#Two years ago,) Szoboszlai achieved that Liverpool is the first’
 (i.e. It’s still his two-year-old achievement that Liverpool has been the first in the last two years.)

⁷While the speech-reporting class seems to homogeneously describe propositional content, the non-speech-reporting class is not homogeneous. According to Tanya Bondarenko (p. c.), while emotive non-speech-reporting clauses like the complements of ‘think’ and ‘believe’ can be described as “the experiencer has a mental state colored by this emotion with content”, content-less emotive states like the complements of ‘see’ and ‘hear’ can be described as “the experiencer feels this emotion because of the situation of this kind”. Interestingly, if we replace ‘think’ in (7) with ‘have a thought’ in (51), the simultaneous reading disappears.

- (51) *Peti-nek az vol-t a gondolat-a, [hogy Mari szomorú vol-t].*
 Peti-DAT it be-PST.3SG the thought-3SG.P that Mari sad be-PST.3SG
 ‘Peti had the thought: “Mari was sad.” but ?? ‘Peti had the thought: “Mari is sad.”’

⁸While these adverbs can also occur post-verbally, linear position does not reflect hierarchical position in the domain c-commanded by the verb (É. Kiss 2008).

⁹The full empirical picture is somewhat more complicated as certain speech-reporting clauses, e.g. the complement of *mond* ‘say’, allow cross-clausal movement to Spec,FocP even though they lack the simultaneous reading. This does not contradict to the main text if speech-reporting clauses can differ in size: While the complements of manner of speech verbs and clauses with high sentential adverbs are SayPs, the complement of *mond* ‘say’ can be α P such that $\text{Say} > \text{Foc} > \alpha > \text{T}$ in *fseq*. Under this analysis, every speech-reporting clause has the α P layer but some may lack the SayP layer. Another thing to note is that

clauses categorized here as SayPs are not islands, e.g. relative movement can proceed from them (Egressy 2023, 2025). This shows that they cannot be analyzed as quotations.

¹⁰ As a third alternative, the selective presence of simultaneous readings in Hungarian can be interpreted in wider context of the diachronic disappearance of *indexical shift*, which is parallel to the appearance and development of finite embedding. According to É. Kiss (2023), Hungarian clauses introduced by *hogy* ‘that’ were originally correlative clauses juxtaposed with the matrix clause. Originally, these juxtaposed *hogy*-clauses exhibited indexical shift for every indexical, however, as the language changed historically from SOV to SVO, the embedded clauses got more and more integrated in and dependent on the matrix clause. As a part of this, Hungarian lost the indexical shift of personal pronouns (É. Kiss 2023). If we chose to analyze past tense as an indexical, it would be possible to say that the independence of past-tense in speech-reporting clauses is a remnant of the general indexical shift. This is in line with the typology of Deal (2020): The shift of personal pronouns implies the shift of temporal indexicals but not the way around and the complements of non-speech verbs are less shifty than the complements of speech verbs.

¹¹ One may ask whether adjuncts are not islands to syntactic operations (as in Huang 1982). Following the idea that Hungarian topics are adjuncts (Szendrői 2003; É. Kiss 2010; Egressy 2025), Egressy (2023) proposes the same αP - αP adjunction for topic clauses as in (41) (Example 120). As Egressy (2023) shows in Example 8, such left-adjoined topic clauses are not islands to movement to positions above the adjunction-site. Cross-linguistically, recent studies revealed that not all adjuncts are islands (e.g. Privoznov 2021).

¹² Recall that the SOT rule applies optionally; if it does not apply, the embedded clause is backstitched (if pragmatics allows it). In (43), if the SOT rule between the intermediate and the deepest Ts/TPs does not apply, the matrix *PAST* may still delete either of these.

¹³ Thanks to Tamari Berulava (p.c.), who mentioned that Georgian may be a language exhibiting a similar size- or clause-type-sensitive SOT pattern to Hungarian.